

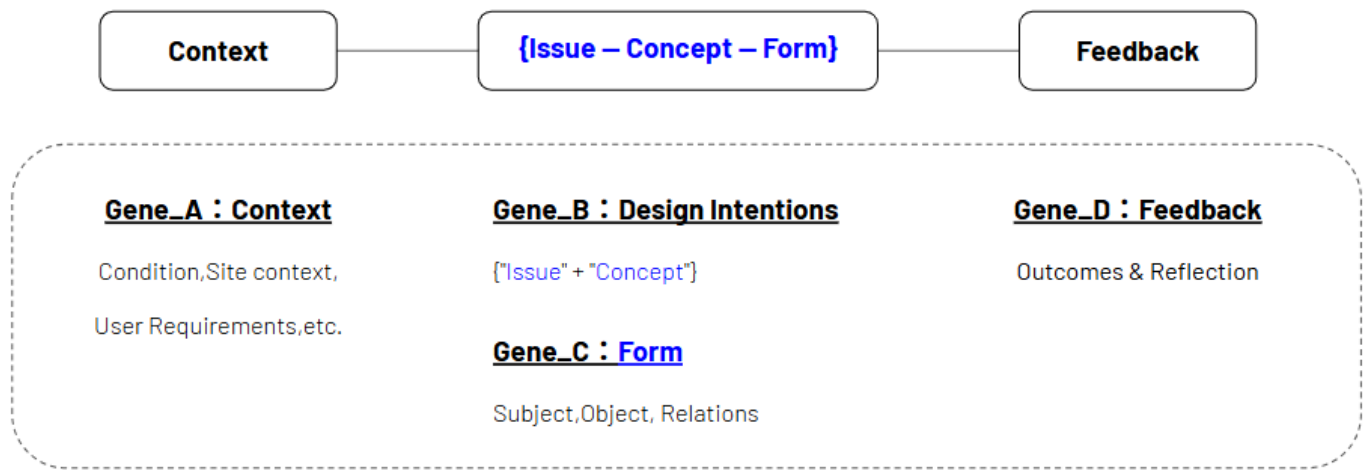
AI in Architecture:

Building a Case-Based Knowledge System and Applying It in Design Practice

Yun-Jhen Lai – Research Portfolio / Advisor: Sheng-Fen Chien

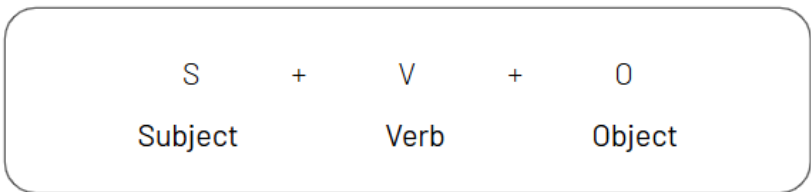
01 Building an AI case-based knowledge system

This study refers to the “Design Story” structure by Oxman & Oxman (1993; 1994), which uses [issue–concept–form] as the basic thinking process. Based on that, I developed a method called “Precedent DNA” to organize case knowledge in architecture. Since LLMs read and process information by tokens, Precedent DNA uses architectural semantic modules instead of long narrative sentences like Design Story.



In the process of building the AI knowledge base, it became clear that LLMs produce different results depending on the type of input data. For example, there is a significant difference between how ChatGPT handles text and how it handles images. Therefore, it is necessary to create specific processing rules for different types of information sources, in order to make the generated text better match the expected outcome.

Story : A spiral ramp surrounding the central atrium



S+V+O
ramp
+
spirals around
+
atrium

In processing architectural case knowledge, a syntax node structure was proposed to improve the efficiency of LLMs when handling sentences. This structure modularizes semantics using a basic [Subject + Verb + Object] format.

The use of syntax nodes makes the overall knowledge structure easier to package into modules. Compared to typical narrative sentences, syntax nodes help LLMs reference and reuse vocabulary across different case modules. This structure also supports the organization of varying sentence patterns and design intents, making the model’s handling of knowledge transfer more logical and consistent.

02 AI Reconfiguration - AI as a consultant

Design Phase: Elderly Housing Types and Floor Allocation
— AI as a Senior Living Consultant

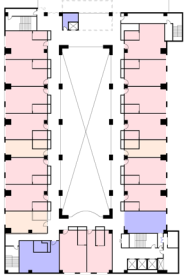
By integrating data such as elderly population statistics, future trends, and Taiwan’s long-term care market analysis, LLMs helped transform this information into a table showing the proportion of elderly by health status. This enables the design to allocate rooms on each floor according to the actual health distribution of seniors, ensuring housing types match their real needs.

Type	Health Status Criteria	Estimate (%)	This Project (%)
Independent Living	No assistance needed for daily activities	45%	46.2% (181 people)
Assisted Living – Light	Needs light help with daily tasks	35%	27.6% (108 people)
Partially Supported Care	Needs moderate help with self-care	15%	19.1% (75 people)
Fully Supported / PAC Care	Needs full-time care or post-hospital rehab	5%	7.1% (28 people)

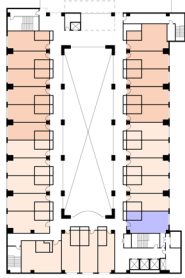
Residentail unit floor plan scale : 1/400



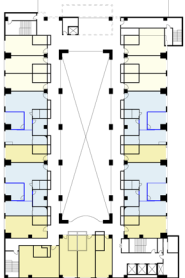
PLAN 8F



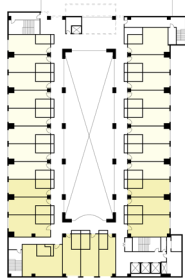
PLAN 9-10F



PLAN 11-13F



PLAN 14-21F



8F	PAC / Intensive Care Floor	3 B1 、 14 C+2 Support Unit
9F–10F	Partial Care Floor	14 B1 、 11 B2 、 1 Support Units
11F–13F	Assisted Living Floor	16 A1 、 10 A2
14F–21F	Independent Living / Family Floor	4D 、 6 A1 、 8 A2

03 AI Reconfiguration - AI as a co-designer

Reconfiguration test.1- Orientation-based Strip zoning
site analysis >>> Orientation-E,S,W,N >>> Congiguration

The Orientation-based Strip Zoning method uses cardinal directions as a primary framework to generate preliminary spatial configurations in response to site-specific conditions.

By aligning architectural programs with environmental and contextual factors—such as functions,sunlight, prevailing winds, and views—this approach produces scaled, data-informed layouts. The AI integrates both qualitative and quantitative insights drawn from a case-based knowledge base to optimize the zoning logic within the given context.

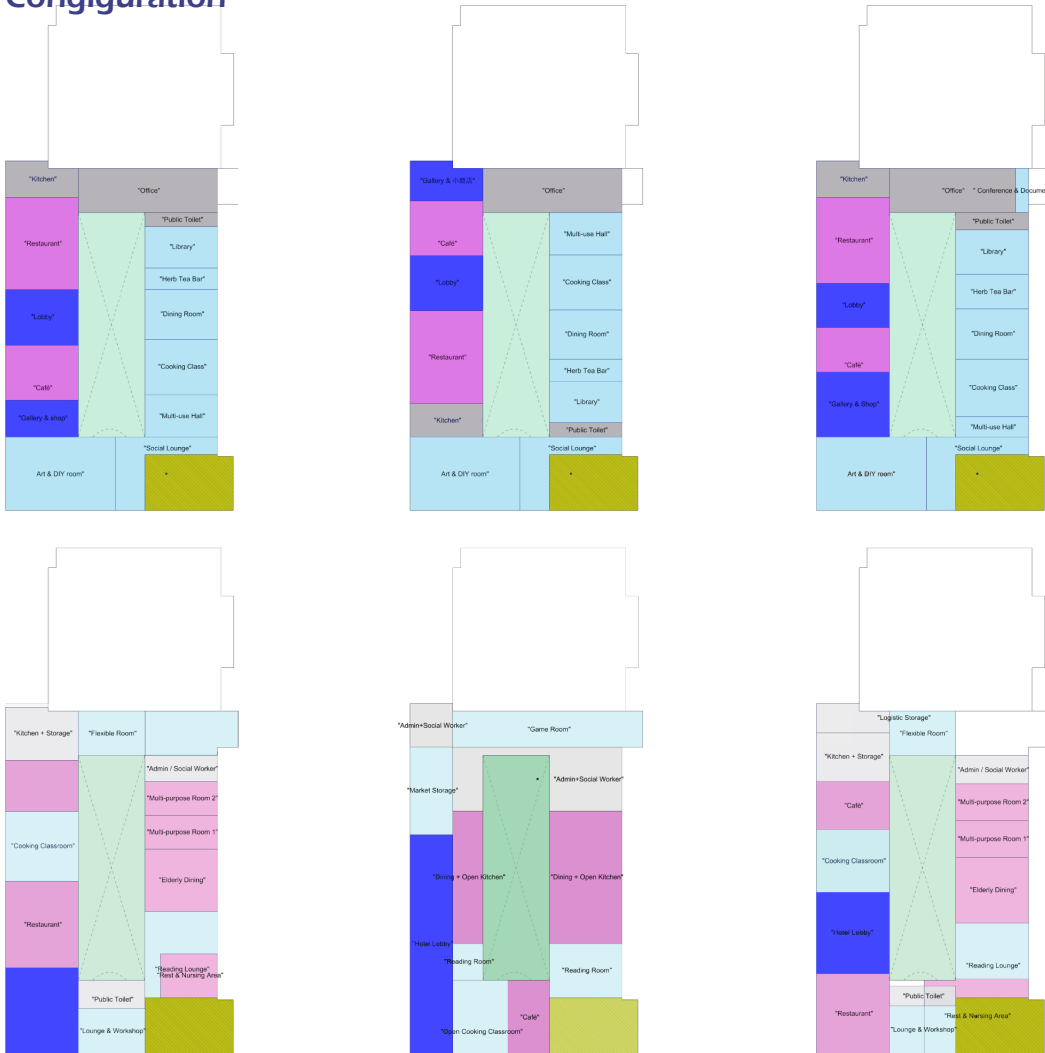
- Public Area

Commercial

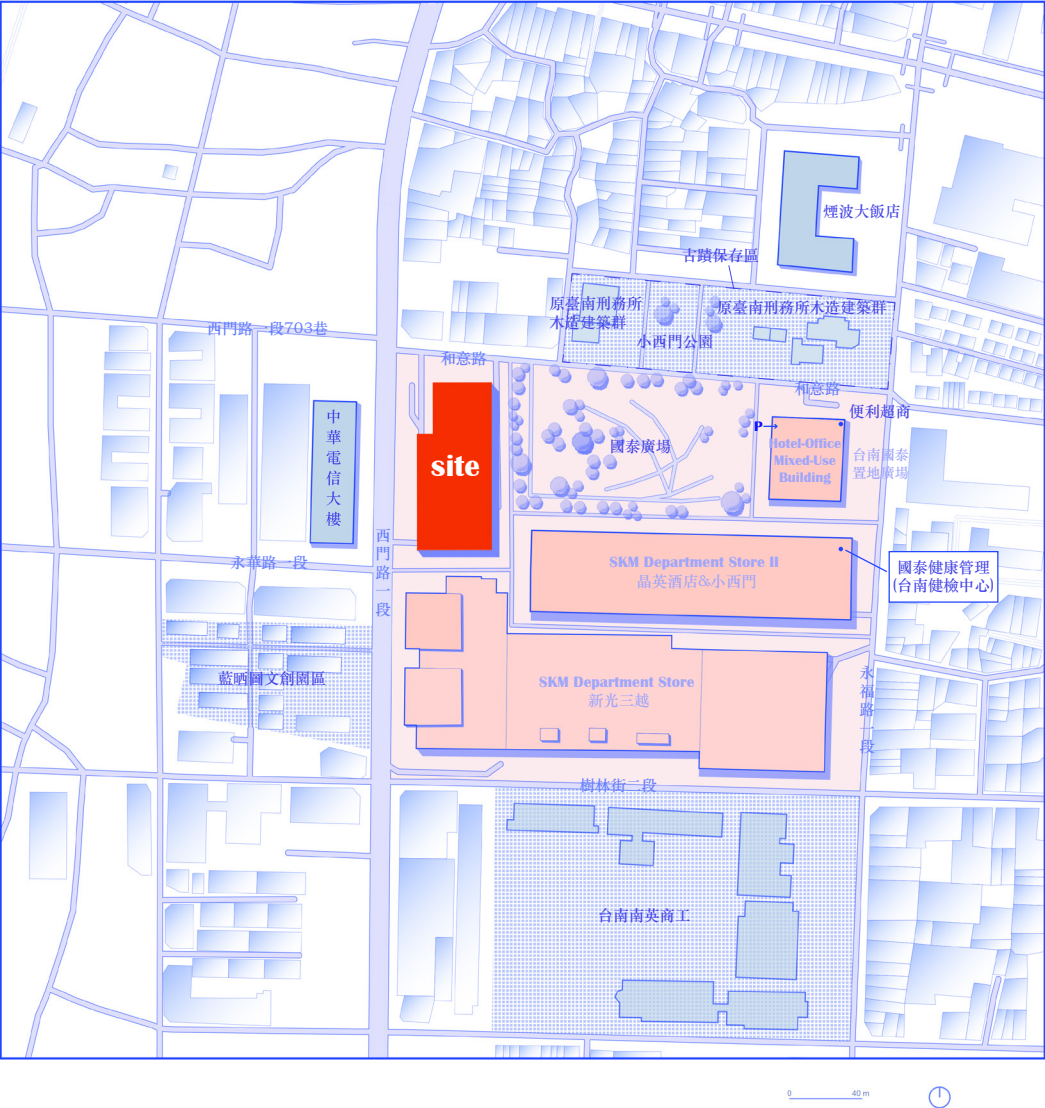
Atrium
- Shared Facility

BOH / Sevice

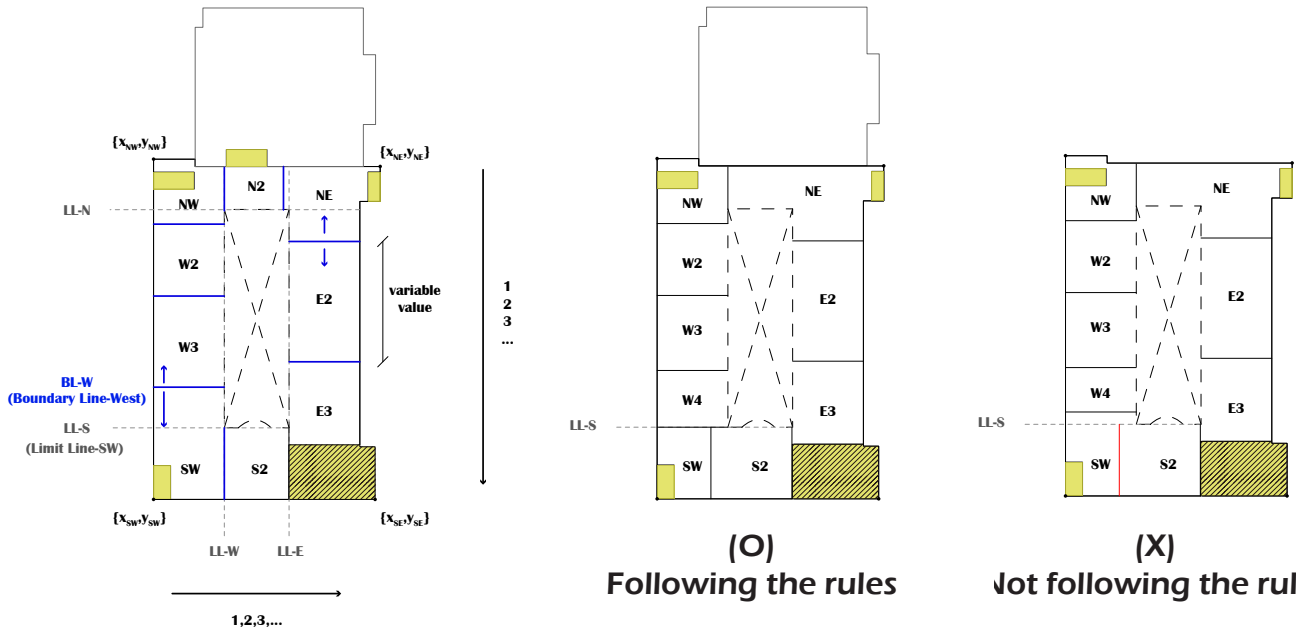
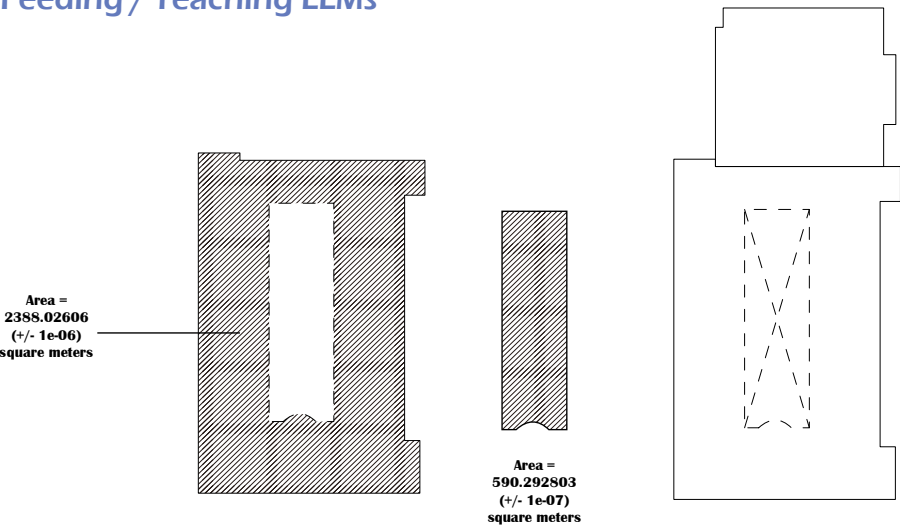
Core



Site Analysis



Feeding / Teaching LLMs



Rules:

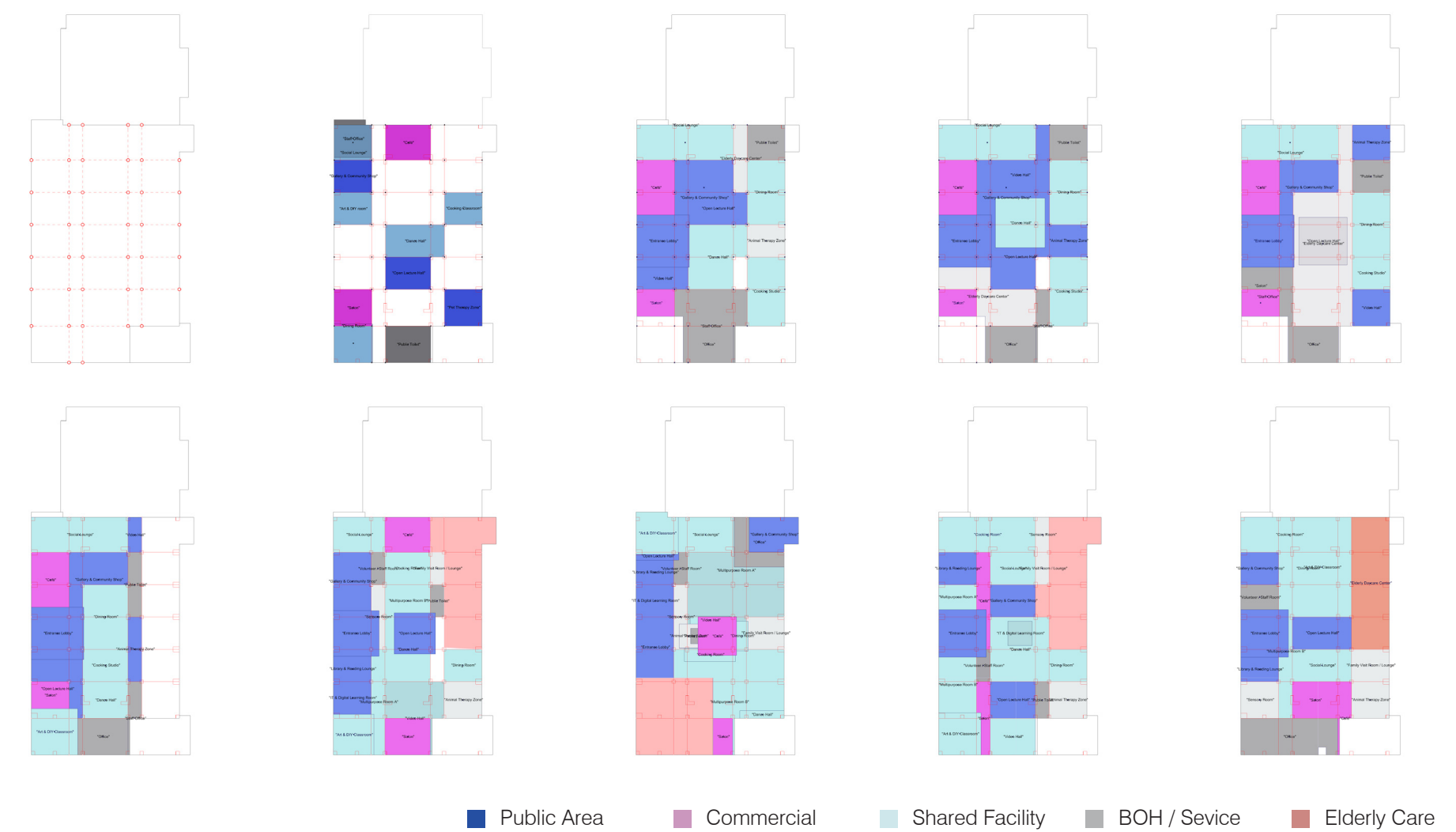
if {y} of BL-W > {y} of LL-S, {x} of BL-S > {x} of LL-W
if {y} of BL-W = {y} of LL-S, {x} of BL-S > {xSW}
if {y} of BL-W < {y} of LL-S, {x} of BL-S = {x} of LL-W

if {x} of BL-S > {x} of LL-W, {y} of BL-S > {y} of LL-S
if {x} of BL-S = {x} of LL-W, {y} of BL-S > {ySW}
if {x} of BL-S < {x} of LL-W, {y} of BL-S = {y} of LL-S

if {x} of BL-N > {x} of LL-W, {y} of BL-S < {y} of LL-N
if {x} of BL-N = {x} of LL-W, {y} of BL-S < {yNW}
if {x} of BL-N < {x} of LL-W, {y} of BL-S = {y} of LL-N

AI Reasoning

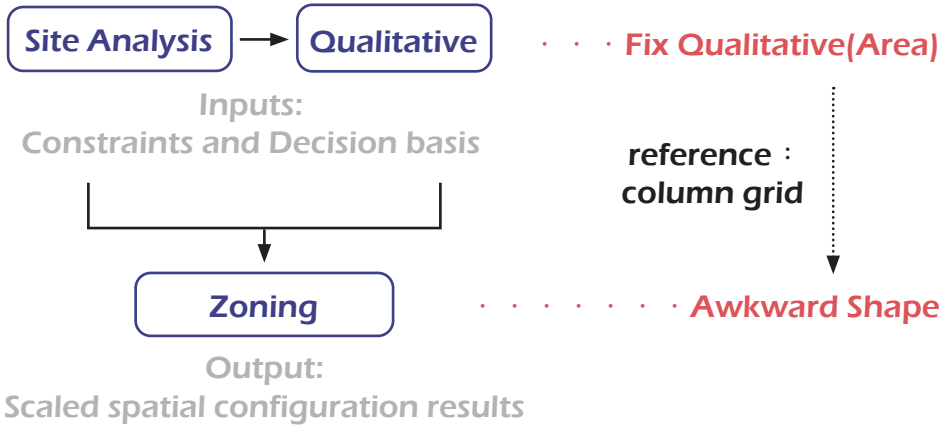
Reconfiguration test.2- Grid Fit Zoning



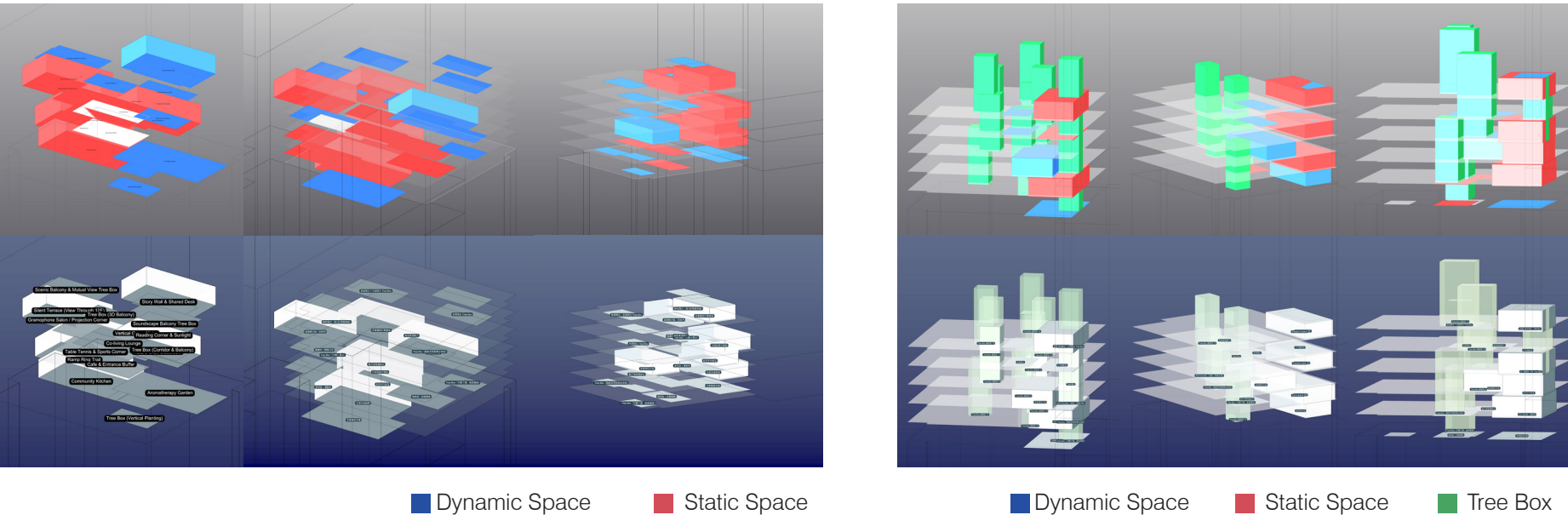
Although the Reconfiguration test.1- **Orientation-based Strip Zoning** method responds well to the site context, the strip zoning is overly uniform and limits the diversity of possible configurations.

Therefore, a second approach, **Grid Fit Zoning**, was tested—using the column grid as a reference for AI to divide the space. Even though the LLM was informed that the grid was **ONLY** a guideline, it still over-adapted to it. Combined with the fixed qualitative and quantitative conditions, this led the model to treat area requirements as rigid constraints. **As a result, the generated layouts had awkward shapes that did not align with geometric intuition or efficient spatial use.**

► Steps in AI reasoning



Reconfiguration test.3
- Semantic Relationship + Spatial Type 3D Zoning



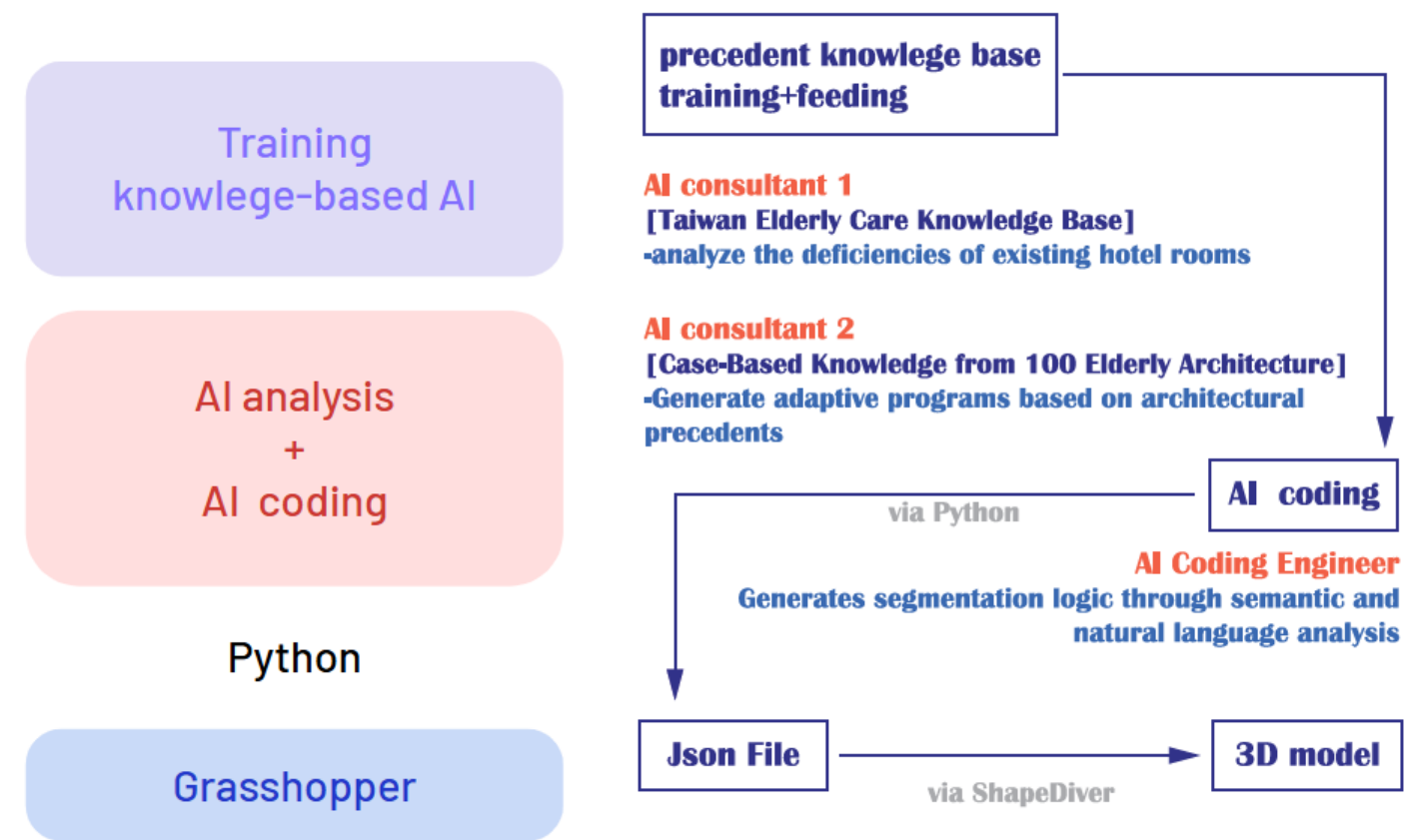
In this stage, I attempted to translate the semantic relationship suggestions generated by LLMs(as a consultant) into three-dimensional spatial forms. Unlike the previous two tests, which were focused on two-dimensional configurations, this step involved defining the actual spatial volumes. This transformation was developed using Grasshopper.

For example, all configurations were represented by sets of points (which can be connected into closed curves). I defined spatial types with specific 3D characteristics:

- 1.Indoor spaces : A one-story-high volume
- 2.Outdoor spaces : a base slab
- 3.Semi-outdoor spaces : a base slab + a top slab.

These definitions resulted in different combinations of components within Grasshopper. The process of converting LLM-generated semantic sentences into data usable in Grasshopper is explained in detail in the next section, titled **"Workflow"**.

04 Designing Workflow



▼ Design Result



Workflow

Part 1. Establishing the AI Consultant

Step 1 — Build the AI knowledge base

Including architectural precedent data, domain-specific knowledge, and design guidelines

Step 2 — Write prompts to instruct the AI

Detailing the reasoning process step-by-step

Step 3 — Define the expected output format

Specify output format and purpose

Part 2. Building the AI Co-designer

Step 1 — Set up all required AI consultants

For this project, the following consultants were created:

AI Consultant 1: Taiwan Elderly Care Knowledge Base

AI Consultant 2: Case-Based Knowledge from 100 Elderly Architecture Projects

Step 2 —Create an AI Coding Engineer

At first, I used the LLM to generate layout data directly, but frequent errors led to the creation of an AI Coding Engineer to handle logic and constraints via Python scripts.

Step 3 — Run Python

To output configuration data(saved as .json)

Step 4 — Import into Grasshopper

Using ShapeDiver's Access JSON component (along with several others), the data is converted into geometric representations.

Step 4 — Bake into Rhino

Final adjustments and refinements are made manually in 3D.

```
{ "8A": {  
  "type": "polyline",  
  "data": {  
    "points": [..... ]  
  }  
},  
"AIRE":"The Added Life"  
.....
```

Access Json

"The Added Life"

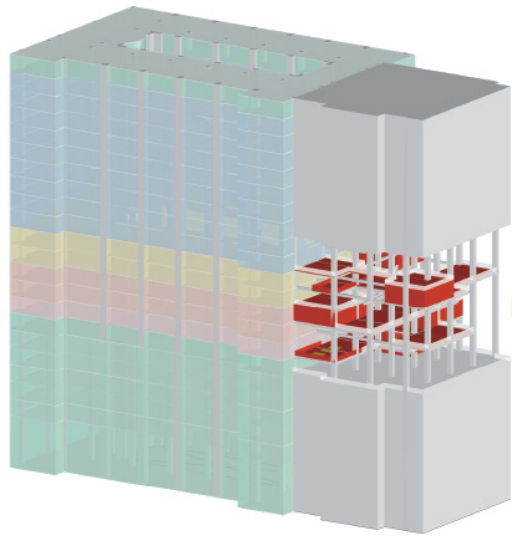
"AIRE"

▲ How "Access Json" component work

▼ Decision basis Example:
A Spatial Quality Evaluation Matrix
(for this case)

9F-10F : 27.03
11F-13F : 29.39
16F,18F-19F : 33.37
17F,20F-21F : 33.97

1-7F
Public Area
/Shared Facilities



各樓層空間品質評估表

樓層	(一)室內品質(m2/人)	(二)公共空間(m2/人)	(三)綠化面積(每人綠化m2)	(四)景觀空間(面積/人)
1 8F	22.03	19.03	0.77	0.45
2 9F	9.55	16.39	0.69	0.0
3 10F	9.55	16.39	0.69	0.0
4 11F	11.56	16.39	0.72	0.72
5 12F	11.56	16.39	0.72	0.72
6 13F	11.56	16.39	0.72	0.72
7 14F	15.95	16.86	0.65	0.51
8 15F	15.95	16.86	0.65	0.51
9 16F	15.95	16.86	0.65	0.51
10 17F	15.95	16.86	0.65	0.51
11 18F	15.95	16.86	0.65	0.51
12 19F	15.95	16.86	0.65	0.51
13 20F	15.95	16.86	0.65	0.51
14 21F	15.95	16.86	0.65	0.51

05 Self Reflection

Part 1 – Building the Precedent Knowledge Base

One of the biggest weaknesses in the early stage was not properly verifying the accuracy of ChatGPT's outputs when building the precedent knowledge base. Every time I tried a new method, I was eager to move forward and kept experimenting without checking whether the AI's content was actually correct. I ended up going deeper and deeper into details based on potentially flawed foundations.

Part 2 – Finding Practical Design Applications

This part was a bit confusing and uncertain for me. For a few weeks, I didn't have a clear direction—I wasn't sure how to develop a method that was not just "conceptually interesting," but actually useful and applicable to real architectural design.

Later, while searching for tools, I found a Grasshopper plugin called ShapeDiver, which luckily gave me a clear entry point. After that, even though I still faced many technical problems, I was able to overcome them with the help of ChatGPT. Sometimes, the roadblocks were caused by ChatGPT—but ironically, the solutions often came from it too.

Summary

Besides gaining a much deeper understanding of LLMs and AI this semester, the most meaningful takeaways for me are:

1. Becoming More Structured in My Thinking

Not just in writing better prompts, but in my overall thinking process. Before, when working toward a goal, I wouldn't break it down into specific steps. But because I had to work with LLMs to reach my design goals this time, I started using Chain-of-Thought prompting and eventually internalized the habit. Now, I naturally try to break tasks into clear, step-by-step plans.

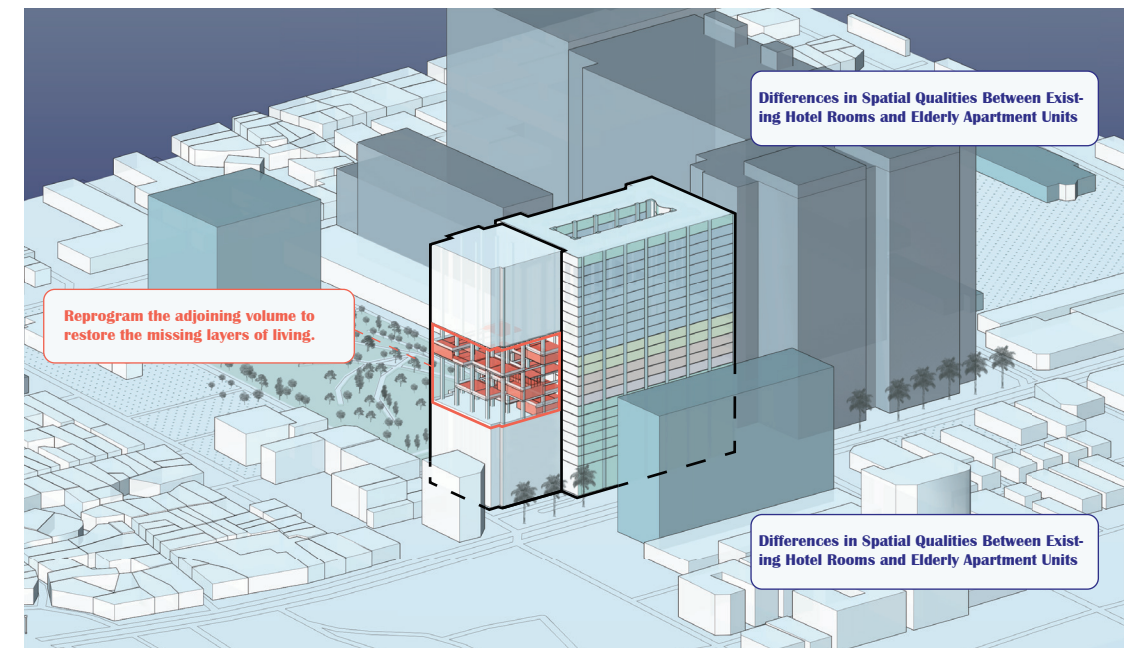
2. Knowing When to Trust (or Not Trust) the AI

LLMs like ChatGPT are great at sounding convincing—even when they're wrong. After interacting with it so much, I've learned that the best way to fact-check is to use Perplexity as a second opinion. It usually gives more accurate responses, plus it shows source links for manual verification.

Although ChatGPT isn't always reliable, I still feel grateful that I didn't give up during the most technical stages like API setup and Python scripting. Even when I got "led astray" by the AI, I just kept patiently debugging with it—and eventually found solutions.

Conclusion:

When the problem involves facts or situations where it's hard to judge right from wrong, don't fully trust the AI. But when the outcome is binary—either it works or it doesn't—then sometimes the best approach is to blindly believe and patiently follow through.



▲ Design Concept

06 Acknowledgements

I would like to express my deepest gratitude to Professor Sheng-Fen Chien for her invaluable guidance and support. She not only provided us with rich AI resources and research funding to explore large language models (LLMs), but also invited experts in the field to share their insights, which broadened my understanding of this emerging discipline. As a student just beginning to explore AI, I am especially thankful for her patient teaching, which gave me a solid foundation and clear direction for my research.

Secondly, I would like to thank my teammate YiXuan Lee, especially for her contributions in the early stage of constructing the precedent knowledge base. Thank you for organizing my initial ideas into a well-structured system and turning it into a paper, which allowed me to learn so much more. I also learned many practical skills for working with LLMs from YiXuan.

I also want to extend my special thanks to Jen-Chien Chien, who expanded the precedent knowledge base to 100 cases, which was extremely helpful for the later stages of my research.

Last but not least, I would like to thank all the members of [GEDEEL 3.0](#). The weekly presentations consistently inspired me and expanded my imagination about what AI can bring to us as architects.

07 References

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Oxman, R., & Oxman, R. (1994). Cognitive models in design case libraries. *Automation in Construction*, 3(2–3), 113–122.